Practical – 11

Practical Title: Student Management CRUD with Search and Filter

Problem Definition: Create a student management module to add, view, edit, delete, search and filter student records. All database operations must use MYSQLi statements and clear success/failure messages.

A. Theory

1. Are SELECT, INSERT, UPDATE, and DELETE operations working?
Yes. All four SQL operations are implemented in the Student Management System.

- SELECT: Retrieves and displays student records from the database.
- INSERT: Adds new student records to the database.
- UPDATE: Modifies the existing details of a student.
- DELETE: Removes a student record from the database.

These operations are performed using PHP MySQLi prepared statements.
2. Is search by name/email/course implemented?
Yes. The search functionality is implemented in index.php.
Users can search for student records using:

- Student name
- Email address
- Course name

A separate course filter is also provided to display students belonging to a selected course.
3. Are delete operations confirmed before execution?
Yes. A JavaScript confirmation dialog is displayed before deleting a student record.
The user must confirm the deletion before the form submits a POST request to delete.php. If the user cancels, the deletion does not take place.

B. Implementation/ Output/Result:

The Student Management System was developed using PHP, MySQL and MySQLi in the XAMPP environment. A database named practical11_db was created with a students table containing student ID, name, email and course details. Twenty sample records were inserted for testing.

The application uses MySQLi to connect PHP with the database. The main page displays student records and provides search functionality by name, email and course, along with a course filter.

Separate PHP files were created to perform CRUD operations. The add.php file inserts new student records, edit.php updates existing records, and delete.php removes selected records. A confirmation dialog is displayed before deletion. Input validation, prepared statements and success or failure messages were also implemented.

Finally, the application was tested to verify all CRUD operations, searching, filtering and deletion confirmation.

C. Evidence of Work (Screenshots):

Student Management System

Add New Student

Search and Filter Students

Search by Name, Email or Course: Filter by Course: All Courses Search Reset

Student Records

ID	Name	Email	Course	Actions
20	Sana Desai	sana@gmail.com	Mechanical Engineering	Edit | Delete
19	Manav Joshi	manav@gmail.com	Computer Engineering	Edit | Delete
18	Riya Mehta	riya@gmail.com	Civil Engineering	Edit | Delete
17	Om Shah	om@gmail.com	Information Technology	Edit | Delete
16	Nisha Patel	nisha@gmail.com	Electrical Engineering	Edit | Delete
15	Yash Desai	yash@gmail.com	Computer Engineering	Edit | Delete
14	Kavya Joshi	kavya@gmail.com	Mechanical Engineering	Edit | Delete
13	Jay Mehta	jay@gmail.com	Civil Engineering	Edit | Delete
12	Ananya Shah	ananya@gmail.com	Computer Engineering	Edit | Delete
11	Rahul Patel	rahul@gmail.com	Information Technology	Edit | Delete
10	Mira Desai	mira@gmail.com	Electrical Engineering	Edit | Delete
9	Dev Joshi	dev@gmail.com	Mechanical Engineering	Edit | Delete
8	Isha Mehta	isha@gmail.com	Computer Engineering	Edit | Delete
7	Arjun Shah	arjun@gmail.com	Civil Engineering	Edit | Delete
6	Neha Patel	neha@gmail.com	Information Technology	Edit | Delete
5	Karan Joshi	karan@gmail.com	Electrical Engineering	Edit | Delete
4	Priya Desai	priya@gmail.com	Computer Engineering	Edit | Delete
3	Rohan Mehta	rohan@gmail.com	Mechanical Engineering	Edit | Delete
2	Diya Shah	diya@gmail.com	Information Technology	Edit | Delete
1	Aarav Patel	aarav@gmail.com	Computer Engineering	Edit | Delete

Fig1: Student Records Display

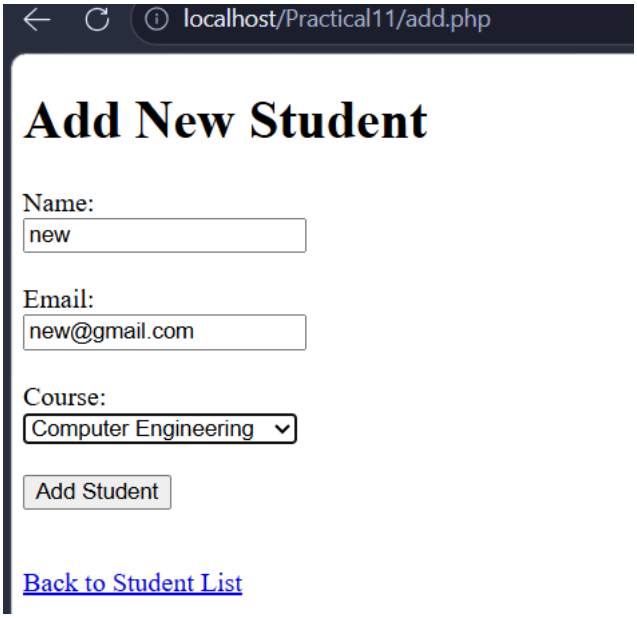


Fig2: Add Student Form

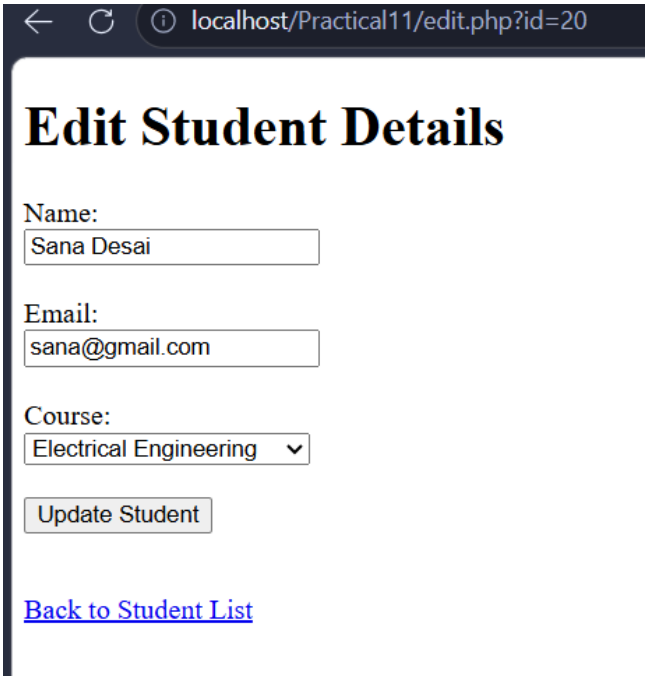


Fig3: Edit Student Details

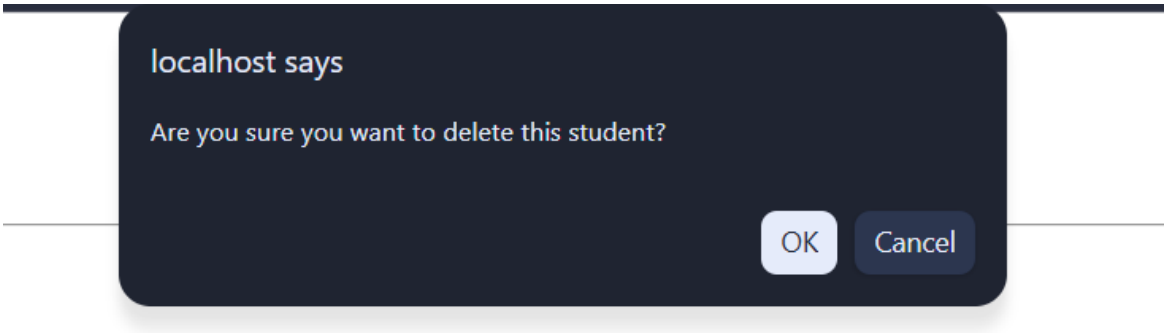


Fig4: Delete Confirmation

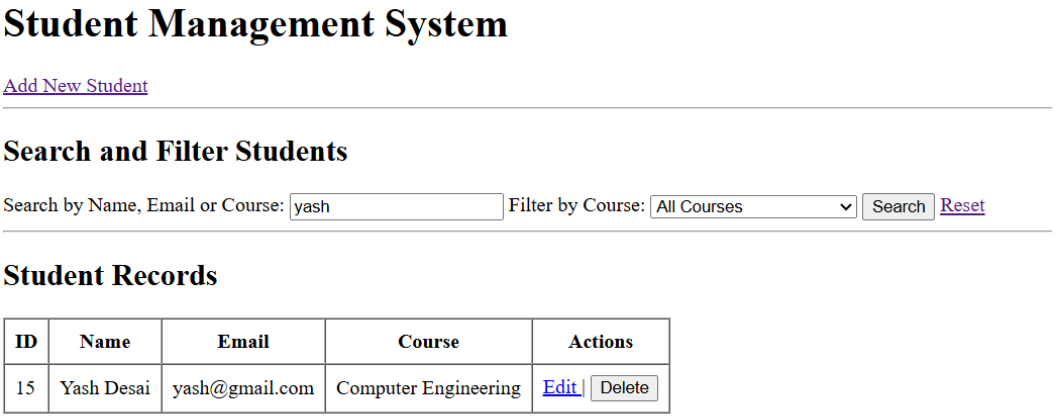


Fig5: Search Functionality

D. GitHub Repository Link:

aryan150806-wq/StudentHubPortal: Web Development Framework P

E. Challenges Faced:

1. Establishing a proper connection between PHP and the MySQL database using MySQLi.
2. Writing SQL queries to perform CRUD operations without errors.
3. Implementing search and course filtering to display the correct student records.
4. Handling input validation, duplicate email addresses, and deletion confirmation.

F. Learning Outcomes:

The Student Management System was successfully developed using PHP and MySQL. The practical helped in understanding database connectivity and implementing CRUD operations using MySQLi. Search and filtering functionalities were also implemented to retrieve specific student records efficiently. Overall, this practical improved understanding of server-side programming and database management.

Faculty Remarks: